

sustainable transportation.

Tranzitor was founded in 2019 in Ahmedabad, Gujarat with a mission to revolutionise the logistics sector and combat the escalating pollution crisis in urban areas. Devesh Patel, the founder of Tranzitor, realised that the diesel-based vehicles used in the logistics sector often emit more pollution than any other vehicles on the road. To address this pressing issue and reduce emissions, he decided to import and sell retrofitment kits for old diesel trucks.

When discussing Tranzitor's journey and motivation to develop indigenous powertrain solutions, Devesh explains, "We initially started with the idea of retrofitting old diesel trucks to electric. However, during this journey, we realised that to ensure a reliable and cost-competitive product, we couldn't depend on sourcing components. Instead, we needed to build crucial components such as battery management systems, motor controllers, and dashboards from the ground up."

Motivated by the Make in India initiative, Tranzitor began developing indigenous solutions such as battery management systems (BMS), foot pedals, motor controllers, etc. Designing and developing these components in-house provided Tranzitor with complete control over the quality, performance, and compatibility of their powertrain solutions. This approach ensured that their products would not only be dependable but also tailor-made for the specific requirements of the Indian market and suitable for Indian weather and road conditions.

One of Tranzitor's latest offerings is the DWI 1.0 motor controller,



*DWI 1.0 motor controller*



*Devesh Patel, Founder and CEO, Tranzitor*

which is specifically designed for high-power applications in two-wheelers on Indian roads. The motor controller can cater to 3kW and 5kW motors. It also offers various safety and smart features, including regenerative braking.

When asked about the considerations for determining the controller's specifications, Devesh explains, "When we embarked on the journey to build a controller, market research indicated a growing demand for high-speed and high-performance two-wheelers. Instead of focusing solely on low-power, low-cost controllers, we aimed to carve out our niche in the realm of

higher power and performance vehicles."

The DWI 1.0 motor controller has an efficiency rating of up to 95%. It offers compatibility with batteries of 48V, 60V, and 72V nominal voltage, making it a versatile choice for various electric vehicle applications. The controller supports both BLDC motors and PMSM motors, with a current handling capacity of 85A and a peak current capability of 300A. With a maximum power delivery of 8kW, the DWI

1.0 motor controller ensures robust output, and for communication, it uses the CAN 2.0. DWI 1.0 motor controller that offers several safety features, including over- and under-voltage protection, as well as over-current protection. It also enables various smart features in the vehicle, such as regenerative braking, hill hold, brake assist, and smooth start, enhancing both comfort and safety during the ride. These advancements by Tranzitor highlight their commitment to developing innovative, indigenous powertrain solutions that cater to the specific demands of the Indian market.

The urgent need for sustainable transportation solutions has brought forth a new era of innovation and technological advancements. By focusing on creating reliable, cost-effective, and high-performance components, companies such as Tranzitor are reshaping the future of transportation. The journey towards sustainable transportation is an ongoing pursuit, recognising the need to transition towards alternative fuels to not only enhance travelling comfort and safety but also to preserve our planet for future generations. **EFY**

—Sharad Bhowmick